

CONTENTS

NOVEMBER 2024 | ISSN-2456-4885

FOR U & ME

- 21 SageMath: A Quick Introduction to Cybersecurity
- 28 India's Leap in Supercomputing: Innovating for Tomorrow

FOCUS

- 42 Containerisation: The Cornerstone of Multi-Cloud and Hybrid Cloud Success
- 50 Exploring eBPF and its Integration with Kubernetes
- 56 Deploying Large Language Models on Kubernetes
- 61 Securing Your Containers: Best Practices for Docker and Kubernetes
- 64 How Do Docker Hub and GitHub Actions Streamline the Continuous Deployment Workflow?
- 69 Leveraging Containers for Deploying Generative AI Applications
- 72 Podman: A Powerful Alternative to Docker

DEVELOPERS

- 78 Helgrind: Detecting Synchronisation Issues in Multithreaded Programs
- 81 The Rest of Rust: A Beginner's Gateway

ADMIN

- 84 Building a Virtual Private Cloud on AWS Cloud
- 91 Fat Containers and Multipass: Building a Cloud on Your Laptop

"Scalability challenges at the edge can be addressed by creating independent failure domains and Kubernetes' declarative infrastructure"

Keith Basil, General Manager
(Edge Business Unit), SUSE



14



18

Efficient Prompt Engineering: Getting the Right Answers



34

**From Virtual Machines to Docker Containers:
The Evolution of Software Development**

REGULAR FEATURES

07 FossBytes